

Two qubit system

$$|\psi_0\rangle = \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} = \alpha_0 |0\rangle + \beta_0 |1\rangle$$

$$|\psi_1\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \alpha_1 |0\rangle + \beta_1 |1\rangle$$

$$|\psi_0\rangle \otimes |\psi_1\rangle = \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} \otimes \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \alpha_0 \alpha_1 \\ \alpha_0 \beta_1 \\ \beta_0 \alpha_1 \\ \beta_0 \beta_1 \end{pmatrix}$$

Two qubit system

$$|\psi_0\rangle = \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} = \alpha_0 |0\rangle + \beta_0 |1\rangle$$

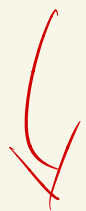
$$|\psi_1\rangle = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \alpha_1 |0\rangle + \beta_1 |1\rangle$$

$$|\psi_0\rangle \otimes |\psi_1\rangle = \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} \otimes \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \alpha_0 \alpha_1 \\ \alpha_0 \beta_1 \\ \beta_0 \alpha_1 \\ \beta_0 \beta_1 \end{pmatrix}$$

$$= \alpha_0 \alpha_1 |00\rangle + \alpha_0 \beta_1 |01\rangle + \beta_0 \alpha_1 |10\rangle + \beta_0 \beta_1 |11\rangle$$

Now imagine n qubits

$$U|\psi\rangle = |\psi'\rangle$$



$$U = \begin{pmatrix} & \end{pmatrix} \begin{matrix} \uparrow \\ 2^n \\ \downarrow \end{matrix}$$

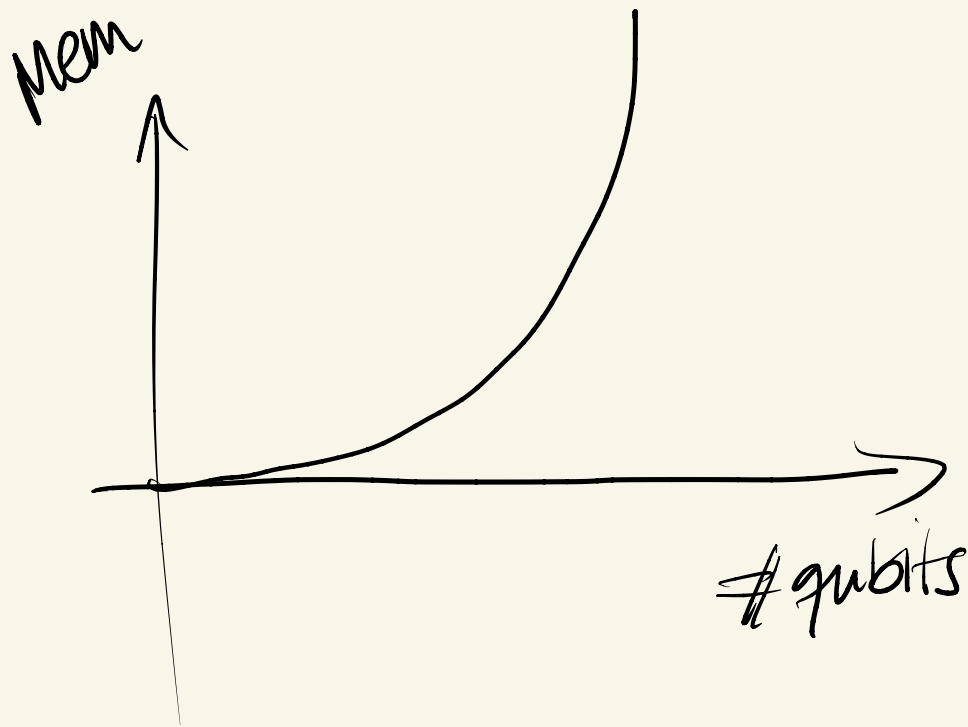
$\leftarrow 2^n \rightarrow$

$$|\psi'\rangle = \begin{pmatrix} \\ \end{pmatrix} \begin{matrix} \uparrow \\ 2^n \\ \downarrow \end{matrix}$$

Amplitudes are complex $\alpha \in \mathbb{C}$

Assume 8 bytes for Real + Complex Part
64 bit floating point

We need $2^n \times 16$ bytes of memory:



1q - 32 B

2q - 64 B

40q - 16 TB

Represent the following two qubit state
in its separable form.

$$|i\rangle \otimes |j\rangle$$

$$|\psi\rangle = \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$

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$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

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$$|0\rangle - \boxed{H}$$

$$|0\rangle - \boxed{X} - \boxed{H}$$

Ex: $|\psi\rangle = \frac{3}{5\sqrt{2}}|00\rangle - \frac{3}{5\sqrt{2}}|01\rangle - \frac{4}{5\sqrt{2}}e^{i\pi/6}|10\rangle$
 $+ \frac{4}{5\sqrt{2}}e^{i\pi/6}|11\rangle$

circuit that prepares $|\psi\rangle$?

$$|\psi\rangle = \frac{3}{5\sqrt{2}}|00\rangle - \frac{3}{5\sqrt{2}}|01\rangle - \frac{4}{5\sqrt{2}}e^{i\pi/6}|10\rangle + \frac{4}{5\sqrt{2}}e^{i\pi/6}|11\rangle$$

Recall the state from last lecture:

$$|\phi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{i\pi/6}|1\rangle$$

what if we do $|\phi\rangle \otimes \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix}$?

$$|\phi\rangle \otimes \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \left(\frac{3}{5}|0\rangle - \frac{4}{5}e^{i\pi/6}|1\rangle \right) \otimes (\alpha_0|0\rangle + \beta_0|1\rangle)$$

$$= \frac{3}{5}\alpha_0|00\rangle + \frac{3}{5}\alpha_1|01\rangle$$

$$- \frac{4}{5}e^{i\pi/6}\alpha_0|10\rangle - \frac{4}{5}e^{i\pi/6}\alpha_1|11\rangle$$

What should be the value
of α_0, β_0 ?

$$|\phi\rangle \otimes \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \left(\frac{3}{5}|0\rangle - \frac{4}{5}e^{i\pi/6}|1\rangle \right) \otimes (\alpha_0|0\rangle + \beta_0|1\rangle)$$

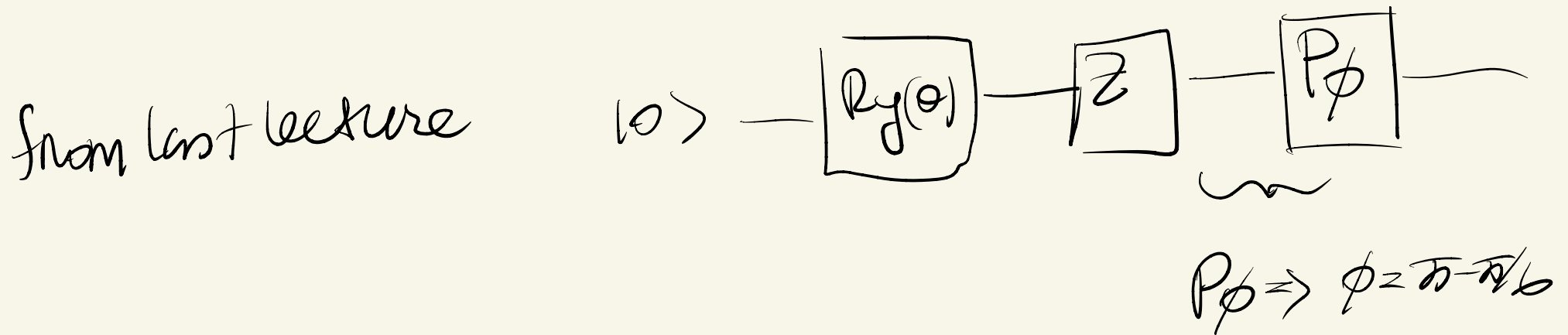
$$= \frac{3}{5}\alpha_0|00\rangle + \frac{3}{5}\alpha_1|01\rangle$$

$$- \frac{4}{5}e^{i\pi/6}\alpha_0|10\rangle - \frac{4}{5}e^{i\pi/6}\alpha_1|11\rangle$$

if $\alpha_0 = \frac{1}{\sqrt{2}}$ and $\alpha_1 = -\frac{1}{\sqrt{2}}$ we get $|\psi\rangle$

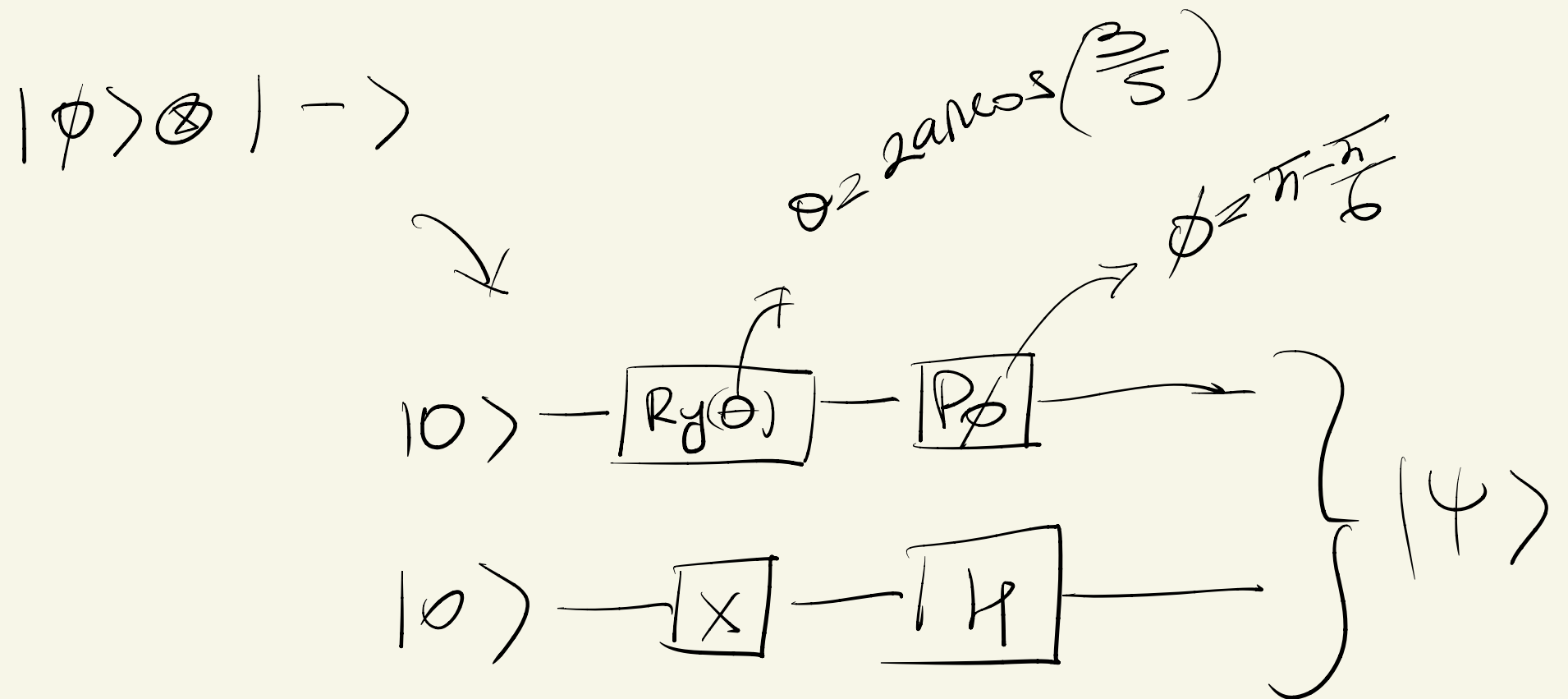
$$\text{So } \begin{pmatrix} \alpha_0 \\ \beta_0 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = |-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

How to prepare $|\phi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{i\pi/6}|1\rangle$?



$$\cos\left(\frac{\theta}{2}\right) = \frac{3}{5} \Rightarrow \theta = 2\arccos\left(\frac{3}{5}\right)$$

Finally, the circuit that prepares $|\Psi\rangle$



Ex 1 Test $U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

on basis states

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$U|00\rangle = |00\rangle$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$U|01\rangle = |01\rangle$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$U|10\rangle = |11\rangle$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

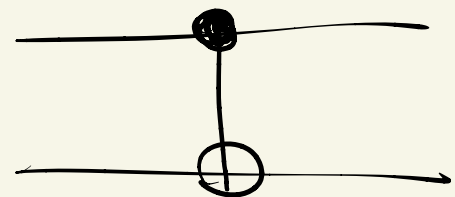
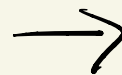
$$U|11\rangle = |10\rangle$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

performs a bitflip
on second qubit

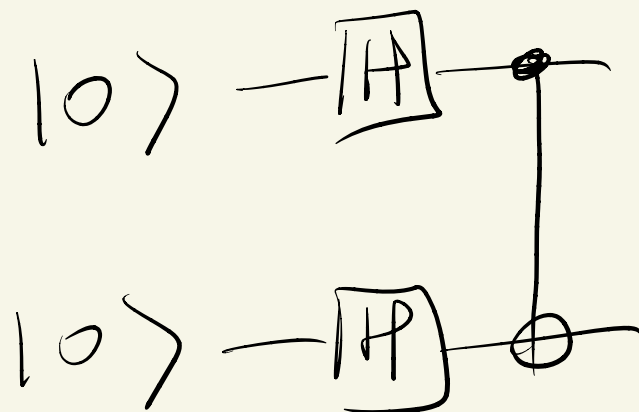
CFF first qubit
 $|1\rangle$

Controlled
NOT

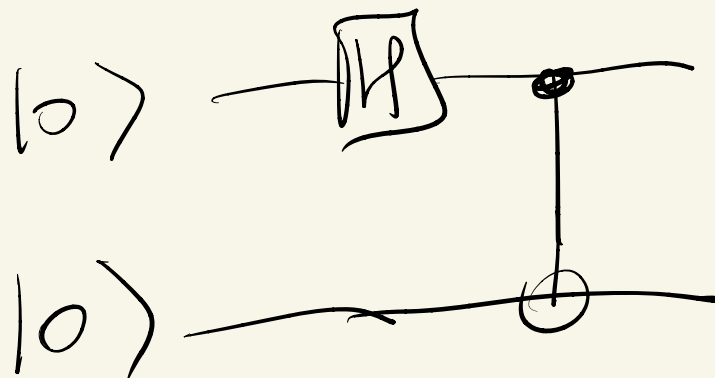


CNOT $|1\rangle$

Ex 2



Ex:



$|4\rangle$?

$$(H \otimes \hat{Z}) |00\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |0\rangle$$

$$= \frac{1}{\sqrt{2}} \underbrace{(|00\rangle + |10\rangle)}_{|\psi'\rangle}$$

$$\text{CNOT } |\psi'\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Represent $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

as $|\psi\rangle \otimes |\psi\rangle$

Represent $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

as $|i\rangle \otimes |j\rangle$

Not possible! we say the state is entangled!

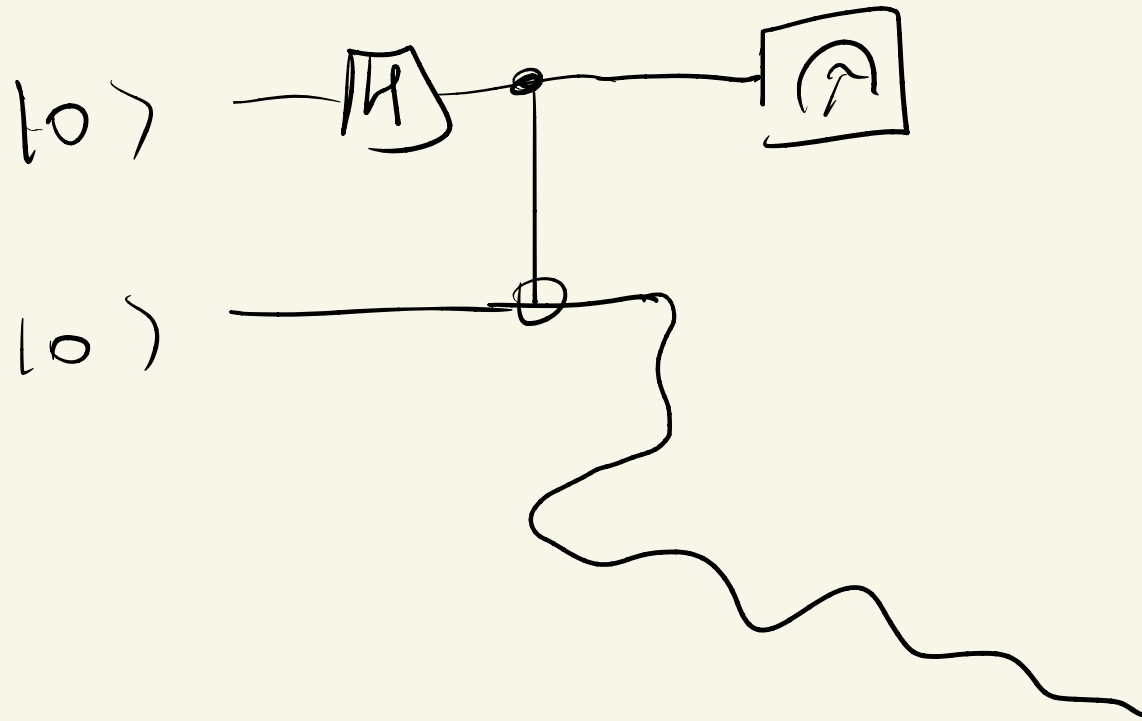
$$\alpha_0 \alpha_1 = 1$$

$$\alpha_0 \beta_1 = 0$$

$$\beta_0 \alpha_1 = 0$$

$$\beta_0 \beta_1 = 1$$

cannot solve for $\alpha_i \beta_i$



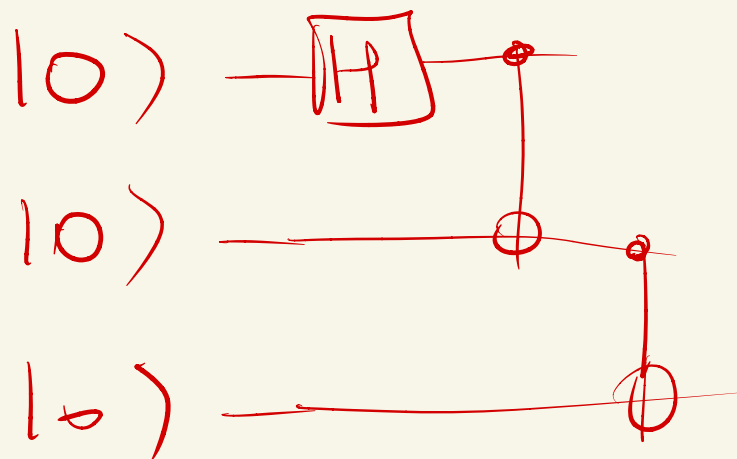
computation on first qubit terminates
we instantaneously know the state
of second qubit !

Ex: Implement the circuit for

$$|4\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

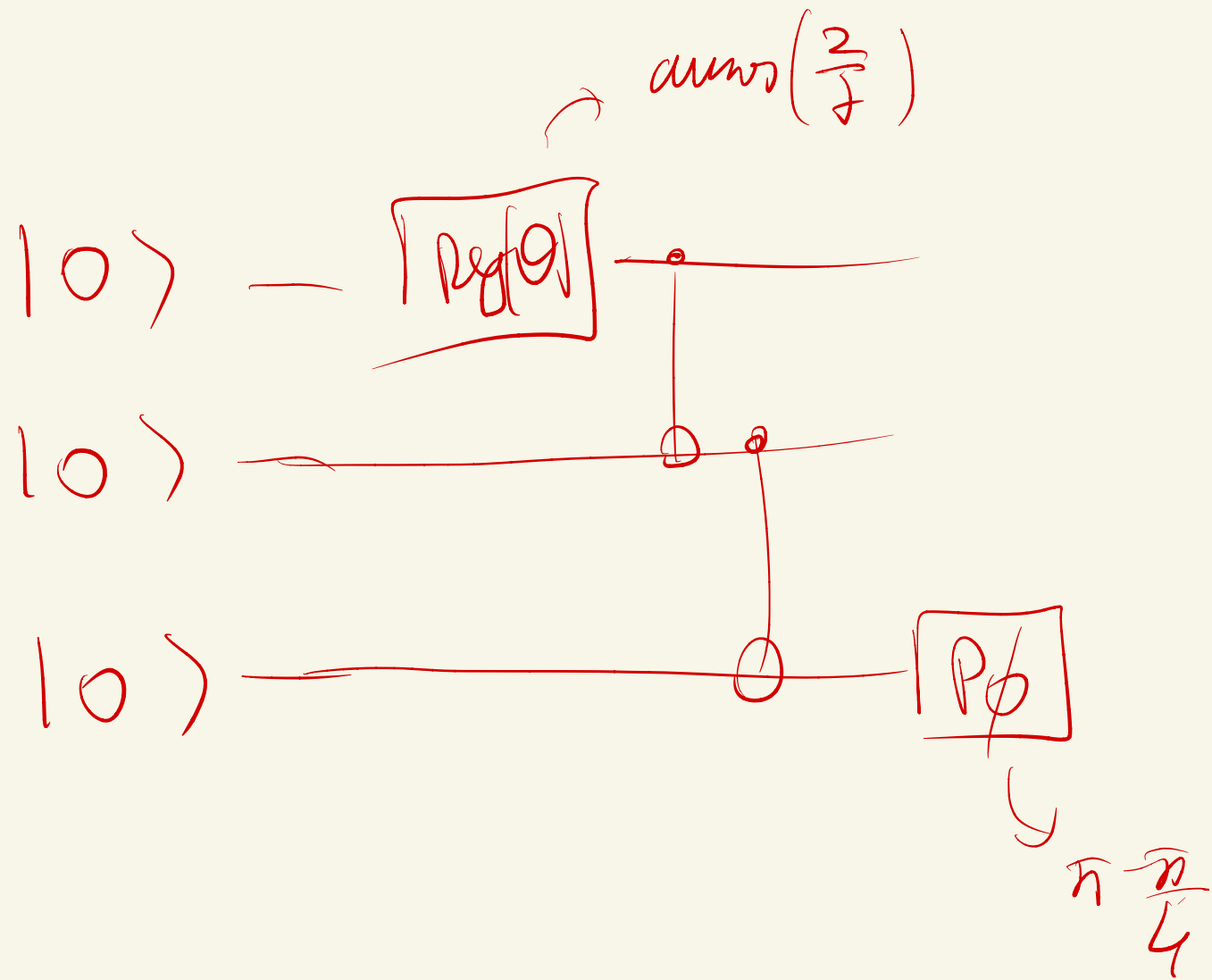
Ex: Implement the circuit for

$$|4\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$



Ex: Complement the circuit for

$$|4\rangle = \frac{2}{7}|000\rangle - \sqrt{\frac{45}{49}} e^{i\pi/4} |111\rangle$$



$$P_{\pi/4} = \boxed{T}$$

gate set so far

$$G = \{H, X, Y, Z, P, R_x, R_y, R_z, \text{CNOT}, T, \dots\}$$

gate set so far

$$G = \{H, X, Y, Z, P, R_x, R_y, R_z, \text{CNOT}, T, \dots\}$$

universal gate set

for quantum computing!

(But there are more universal gate sets)
e.g. $R_x, R_y, R_z + \text{CNOT}$